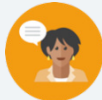


## 1.12 Angle Bisectors

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                          |                                                                                                              |                                                                                                                                                                                                                |                                                                                      |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.                                                                                                                   |                                                                                                              |  Learning Target                                                                                                            | <ul style="list-style-type: none"><li>I can construct an angle bisector.</li></ul>   |
|  Benchmark Coherence | Building On<br>N/A                                                                                                                                                                                                                       |                                                                                                              | Working Toward<br>N/A                                                                                                                                                                                          |                                                                                      |
|  Essential Questions | <ul style="list-style-type: none"><li>What is an angle bisector?</li><li>How can an angle bisector be used to determine measures?</li></ul>                                                                                              |                                                                                                              |                                                                                                                                                                                                                |                                                                                      |
|  Lesson Narrative    | This lesson explores what an angle bisector is and how to construct one using a protractor, patty paper, or compass and straightedge. Students will solve for variables and angle measures using their understanding of angle bisectors. |                                                                                                              |                                                                                                                                                                                                                |                                                                                      |
|  Component Summary | 1.12.1 Warm-Up (~5 min.)                                                                                                                                                                                                                 | Explain agreement or disagreement with student statements ( <i>SE p. 75</i> )                                | 1.12.3 Guided Instruction (15 min.)                                                                                                                                                                            | Solving for variables and angle measures using an angle bisector ( <i>SE p. 78</i> ) |
|                                                                                                       | 1.12.2 Collaboration (20-25 min.)                                                                                                                                                                                                        | Constructing an angle bisector ( <i>SE p. 76</i> )                                                           | 1.12.4 Wrap-Up (~5 min.)                                                                                                                                                                                       | Constructing an angle bisector ( <i>SE p. 79</i> )                                   |
|  Resources         | Glossary                                                                                                                                                                                                                                 |                                                                                                              | Additional Preparation                                                                                                                                                                                         |                                                                                      |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Angle Bisector</li></ul><br><u>Additional Terms:</u> N/A                                                                                                                         |                                                                                                              | <ul style="list-style-type: none"><li>Consider making large copies of the construction steps that can be moved around and put in order on the board during the whole-class discussion of the answer.</li></ul> |                                                                                      |
|                                                                                                       |                                                                                                                                                                                                                                          | Materials                                                                                                    |                                                                                                                                                                                                                |                                                                                      |
|                                                                                                       |                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>Protractor</li><li>Compass</li><li>Ruler</li><li>Patty Paper</li></ul> |                                                                                                                                                                                                                |                                                                                      |

|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may misread the protractor during 1.12.2 Collaboration.</li><li>Students may not open the compass width wide enough during the construction.</li></ul>                                                                                                                                                                                                           | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider how to help students remember the different steps to the constructions in the unit. Students may easily forget steps if they do not understand the whys, the geometric definitions, or the ideas that resulted from the constructions. Consider asking students to write down the four different constructions in the unit and what they remember about each. Use this information to support student learning as they prepare for any unit assessments.</li></ul> |
|                                                                                                   | <b>Literacy and Language Routines</b><br><b>Discussion Supports</b> <p>During 1.12.2 Collaboration, students will be able to discuss and defend a variety of approaches for finding an angle bisector. It may be interesting to ask students to have a mini debate regarding which method is best. Do not clarify if best means most efficient or most accurate; allow students to discuss the merits of each method.</p>                    | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.1.1: Participate in Effortful Learning</b> <p>Starting with the 1.12.2 Collaboration, students are provided with a number of opportunities to discuss and explore what an angle bisector is. Students should be encouraged to work with a partner and discuss options and other ways to explore the angle bisectors.</p>                                                                                                                                         |
|                                                                                                   | <br>Differentiate Instruction <p>Using the strategy of a silent board game during 1.12.2 Collaboration question 4 allows for different entry-points for learners. Students choose when to go to the board to make a move or give an answer based on when they feel comfortable with their knowledge, instead of being selected based on other criteria.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Lesson Notes:                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

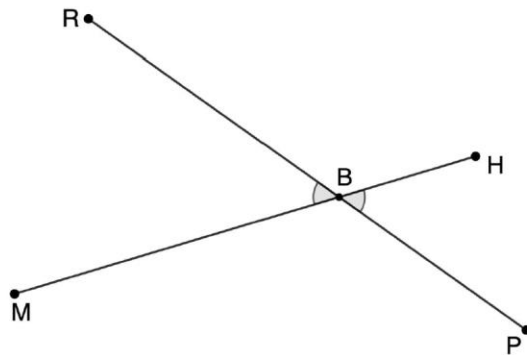
## 1.12.1 Warm-Up: Agree or Disagree

**Component Overview:** Students will engage in an agree or disagree activity where they will review the statements of two students to determine who they agree with or disagree with.



### 1.12.1 Warm-Up: Agree or Disagree

1. Line segments  $MH$  and  $RP$  intersect at point  $B$ .



For this activity, consider having students move to one side of the room if they agree with Jamessia, and the other side with Malik. If they don't agree with either, have them remain seated. Allow students to share out their rationale.

- a. Jamessia says  $\angle MBR$  and  $\angle PBH$  have the same measure. Malik disagrees and says that obviously the measure of  $\angle MBR$  is larger.

Where do you stand?

- ☐ I agree with Jamessia.
- ☐ I agree with Malik.
- ☐ I disagree with both students.

- b. Explain your choice for Part A.

*Answers will vary.  $\angle MBR$  and  $\angle PBH$  are vertical angles so they must have equal measure.*

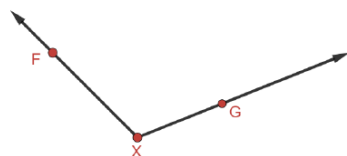
## 1.12.2 Collaboration: Bisecting an Angle

**Component Overview:** Students will find the bisector of an angle using a protractor and using patty paper. Then, the students will construct the bisector of an angle using a compass and straightedge.

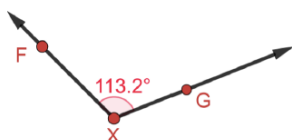


### 1.12.2 Collaboration: Bisecting an Angle

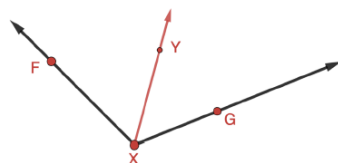
1.  $\angle FXG$  is shown.



- a. Measure the angle with a protractor.  
*Answers will vary. Answers should be close to  $113^\circ$  but do not have to be exact.*

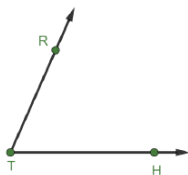


- b. Draw  $\overrightarrow{XY}$  that bisects  $\angle FXG$ .



- c. Discuss with your partner why bisecting  $\angle FXG$  using a protractor and a ruler may not be accurate. Summarize your discussion.  
*Answers will vary.*

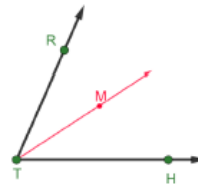
2.  $\angle RTH$  is shown.



*This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed collaboration.*

## 1.12.2 Collaboration: Bisecting an Angle (continued)

- Use patty paper to copy  $\angle RTH$ .
- Fold the patty paper so that  $\overrightarrow{TR}$  and  $\overrightarrow{TH}$  lie on each other. Open the paper and draw  $\overrightarrow{TM}$  along the crease.



- Explain how folding the patty paper bisects the angle.

*Answers will vary. When  $\overrightarrow{TR}$  and  $\overrightarrow{TH}$  line up, the distance from the fold is the same, so the angles are equal. The angles being equal is what makes  $\overrightarrow{TM}$  a bisector of the angle.*



After question 2c, consider asking the following:

- What transformation is represented in this step?
- What maintains congruence when using this transformation?

Another way to more accurately bisect an angle is to use a compass and a ruler.

- $\angle PGW$  is shown.

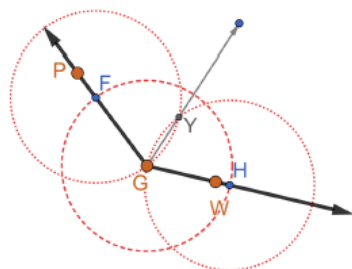
Work with your partner and use the steps below to bisect  $\angle PGW$ .

- Place the compass' point on point  $G$ .
- Adjust the compass' width to any width that will stay on the angle.
- Use the setting to draw an arc that extends across each leg of  $\angle PGW$ . Label the intersections of the arc and the two sides of the angle points  $F$  and  $H$ .
- Place the compass' point on point  $F$ . Keep the width of the compass the same or open the compass wider than the width used to draw the arc in the previous step. Draw an arc in the interior of the angle.
- Without changing the compass' setting, place the compass' point on point  $H$  and make an arc in the interior of the angle. The two small arcs in the interior of  $\angle PGW$  should be intersecting. If they are not intersecting, adjust the compass to a larger width and draw new arcs from points  $F$  and  $H$ .



## 1.12.2 Collaboration: Bisecting an Angle (continued)

- Draw and label point  $Y$  where the two arcs intersect.
- Use a straightedge to draw a ray from point  $G$  through point  $Y$ .



$\overrightarrow{GY}$  bisects  $\angle PGW$ .

4. Olivia found some steps for bisecting an angle on the internet. She wrote them down and then mixed them up to test herself. Her mixed-up steps are shown. Work with your partner to order the steps 1-9.

|   |                                                                                                                                                                     |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9 | Draw a ray from the vertex of the angle to the intersection of the two small arcs.                                                                                  |
| 2 | Stretch the compass to any length that stays on the angle.                                                                                                          |
| 7 | Without changing the span on the compass, place the point of the compass on the other point made by the arc and the other side of the angle and make a similar arc. |
| 4 | Place the compass point on one of the two new points made by the arc.                                                                                               |
| 5 | Change the compass width so that the pencil stays between the rays of the angle.                                                                                    |
| 1 | Place compass point on the vertex of the angle.                                                                                                                     |
| 3 | Swing the compass to make an arc that crosses both rays of the angle.                                                                                               |
| 6 | Draw an arc in the interior of the angle.                                                                                                                           |
| 8 | Mark the point where the two small arcs in the interior of the angle intersect.                                                                                     |



After partners have worked to order the steps, consider implementing a silent board game. With large copies of the steps prepared ahead of time, have students volunteer to come to the board to place the steps in order. Each student will move only one step at a time. Continue allowing students to come until all steps are in order and all agree on the steps. There is no talking during this activity. This activity provides different entry-points, as students volunteer when they know a particular step even if they do not know all of the steps. Using this teaching strategy now will set up expectations if it is used in other lessons, particularly when writing proofs.

## 1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

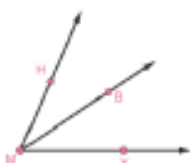
**Component Overview:** Students will use the information about angle bisectors to solve for unknown variables and angle measures. Students will also examine the thinking of two sample students' work and write a note to the student who made a mistake.



### 1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

An **angle bisector** is a line, ray, or line segment that cuts an angle into two equal parts.

1.  $\angle HMX$  is shown where  $\overrightarrow{MB}$  bisects  $\angle HMX$ .

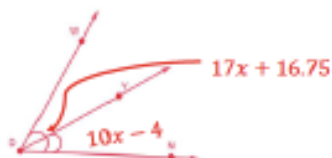


If  $m\angle HMX = 7x + 3$  and  $m\angle BMX = 3x + 6$ , then find the value of  $x$ .

$x = 9$

2.  $\overrightarrow{DY}$  bisects  $\angle WDN$  such that  $m\angle YDN = 10x - 4$  and  $m\angle WDN = 17x + 16.75$ .

- a. Draw a diagram of the description.



- b. Yordis and Hae-Won used the given information to determine the value of  $x$ . Their equations are shown.

| Yordis                               | Hae-Won                              |
|--------------------------------------|--------------------------------------|
| $10x - 4 = \frac{1}{2}(17x + 16.75)$ | $\frac{1}{2}(10x - 4) = 17x + 16.75$ |

- c. Discuss with your partner which student made a mistake setting up the equations.

Answers will vary. Hae-Won should multiply  $10x - 4$  by 2 not  $\frac{1}{2}$ .



Some students may benefit from a discussion on the difference between  $m\angle HMX$  and  $\angle HMX$ . It may also be helpful for students to know that the ray  $\overrightarrow{MB}$  and line segment  $\overline{MB}$  can both be considered the bisector.

## 1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems (continued)

- d. Write a short note to the student who made a mistake in setting up the equations to explain their error to them.

**Means**  
**I**  
**Start**  
**To**  
**Acquire**  
**Knowledge**  
**Experience**  
**Skills**

*Answers will vary.*

3.  $\overrightarrow{BT}$  bisects  $\angle HBN$  such that  $m\angle HBN = 10x - 2$ ,  $m\angle HBT = 5x - 1$ , and  $m\angle TBN = 7x - 23$ . Find  $m\angle TBN$ .  
 *$m\angle TBN = 54$*



*Before moving to question 2d, have students share a summary of their discussion. Encourage students to talk about which angle is the larger and which angle is smaller. Ask how multiplying by  $\frac{1}{2}$  affects the value of a number. When a student gives a valid explanation of which student is correct, have another student repeat what was said. Doing this here will help all students organize their thoughts to write an explanation of the error in the following question.*



*Question 3 can be solved in several ways. Consider releasing students to try this on their own. Monitor for the strategies used and call on students to share their thinking with the group as an opportunity to highlight different ways of thinking about the same problem.*



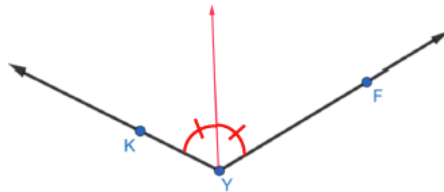
## 1.12.4 Wrap-Up: Ticket Out the Door

**Component Overview:** Students will construct the bisector of an angle.



### 1.12.4 Wrap-Up: Ticket Out the Door

1. Construct the angle bisector of  $\angle KYF$  using a compass and a straightedge.



Consider allowing students to construct the angle bisector or to explain how to construct the angle bisector. Both options provide information about what students understand from the lesson.